

Smart Internship Management System

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Milestone Tracking:

Milestones	Date	Version	Remarks
ERD Diagram and Schema	24/4/2026	1.0	
ERD Diagram and Schema (Updated)	10/5/2026	1.1	
Applied 1NF, 2NF, and 3NF normalization and updated ERD	16/5/2026	2.0	

Milestone 2 — ERD Design & Normalization

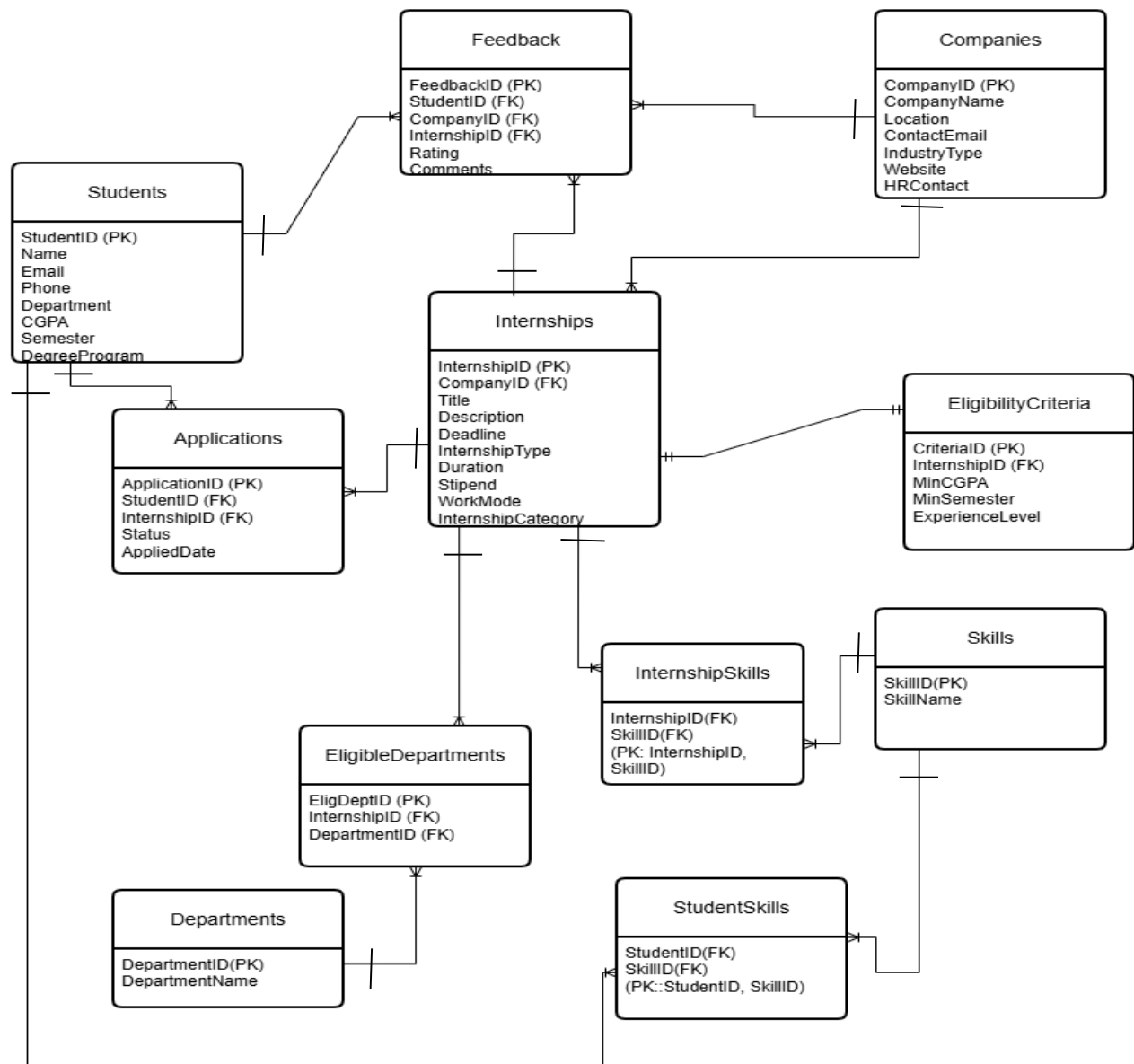
Project Title: Smart Internship Management System

Objective: Normalize the existing database schema from 1NF to 3NF, justify all changes, update the ERD, and prepare the repository for GitHub submission.

Introduction

This milestone focuses on improving the initial database schema of the Smart Internship Management System through formal normalization. The schema was analyzed according to First Normal Form (1NF), Second Normal Form (2NF), and Third Normal Form (3NF). Redundant data, repeated attributes, and dependency issues were identified and corrected. The ERD was also updated to reflect the normalized structure.

Updated Normalized ERD



Normalization Process

First Normal Form (1NF)

A table satisfies 1NF when all attributes contain atomic values and there are no repeating groups or multi-valued attributes.

- **Students:** Already satisfied 1NF because all attributes contain single atomic values.
- **Companies:** Already satisfied 1NF because each field stores only one value.
- **Internships:** Already satisfied 1NF because internship details are stored atomically.
- **Applications:** Already satisfied 1NF because each application is uniquely identifiable and contains no repeating groups.
- **Feedback:** Already satisfied 1NF because all attributes contain atomic values.
- **EligibilityCriteria:** RequiredSkills violated 1NF because multiple skills could be stored inside a single field. This issue was resolved by creating Skills and InternshipSkills tables.
- **EligibleDepartments:** Department names were initially stored as text values. This structure was later improved during 3NF normalization.

Changes Applied in 1NF

- Created Skills table.
- Created InternshipSkills junction table.
- Removed multi-valued RequiredSkills attribute.

Second Normal Form (2NF)

A table satisfies 2NF when it is already in 1NF and all non-key attributes are fully dependent on the entire primary key.

- **Students:** Satisfied 2NF because all attributes depend entirely on StudentID.
- **Companies:** Satisfied 2NF because all attributes depend on CompanyID.

- **Internships:** Satisfied 2NF because all internship attributes depend on InternshipID.
- **Applications:** Satisfied 2NF because all non-key attributes depend completely on ApplicationID.
- **Feedback:** Satisfied 2NF because all feedback details depend on FeedbackID.
- **InternshipSkills:** Satisfied 2NF because the relationship depends on both InternshipID and SkillID.

Changes Applied in 2NF

- Added UNIQUE(StudentID, InternshipID) constraint to prevent duplicate applications.
- Used junction tables for many-to-many relationships.
- Ensured complete dependency on primary keys.

Third Normal Form (3NF)

A table satisfies 3NF when it is already in 2NF and contains no transitive dependencies.

- **Students:** Satisfied 3NF because there are no transitive dependencies among attributes.
- **Companies:** Satisfied 3NF because all attributes directly depend on CompanyID.
- **Internships:** Satisfied 3NF because all attributes directly depend on InternshipID.
- **EligibleDepartments:** DepartmentName caused redundancy and inconsistency. A separate Departments table was created and DepartmentID was used as a foreign key.
- **Feedback:** InternshipID was added to remove ambiguity regarding which internship the feedback belongs to.

Changes Applied in 3NF

- Created Departments table.
- Replaced DepartmentName with DepartmentID in EligibleDepartments.

- Added InternshipID in Feedback table.
- Reduced redundancy and improved consistency.

New Normalized Tables

Departments

- DepartmentID (PK)
- DepartmentName

Skills

- SkillID (PK)
- SkillName

InternshipSkills

- InternshipID (FK)
- SkillID (FK)
- Composite Primary Key: InternshipID + SkillID

StudentSkills

- StudentID (FK)
- SkillID (FK)
- Composite Primary Key: StudentID + SkillID

Duplicate and Redundancy Removal

Repeated department names and multiple skill values were identified as redundancy issues within the original schema. Normalization removed these issues by creating dedicated Departments and Skills tables.

Junction tables were introduced to properly manage many-to-many relationships and eliminate duplicated data.

Updated Relationships

- Students (1) — (M) Applications
- Internships (1) — (M) Applications
- Companies (1) — (M) Internships
- Students (1) — (M) Feedback
- Companies (1) — (M) Feedback

- Internships (1) — (1) EligibilityCriteria
- Internships (1) — (M) EligibleDepartments
- Departments (1) — (M) EligibleDepartments
- Internships (1) — (M) InternshipSkills
- Skills (1) — (M) InternshipSkills
- Students (1) — (M) StudentSkills
- Skills (1) — (M) StudentSkills

Conclusion

The Smart Internship Management System database schema was successfully normalized up to Third Normal Form (3NF). The updated schema eliminates redundancy, improves consistency, and provides a scalable and maintainable database structure. The ERD was updated to accurately represent the normalized schema and all relationships.